Chapter 9

Method of Moments

9.1 Introduction

The approach known as the *method of moments* is described in this chapter. It produces an estimator that is easy to determine and simple to implement. Although the estimator *has no optimality properties*, it is useful if the data record is long enough. This is because the method of moments estimator is usually consistent. If the performance is not satisfactory, then it can be used as an initial estimate, which is subsequently improved through a Newton-Raphson implementation of the MLE. After obtaining the method of moments estimator, we describe some approximate approaches for analyzing its statistical performance. The techniques are general enough to be used to evaluate the performance of many other estimators and are therefore useful in their own right.

9.2 Summary

The method of moments approach to estimation is illustrated in Section 9.3 with some examples. In general, the estimator is given by (9.11) for a vector parameter. The performance of the method of moments estimator may be partially characterized by the approximate mean of (9.15) and the approximate variance of (9.16). Also, any estimator that depends on a group of statistics whose PDF is concentrated about its mean can make use of the same expressions. Another case where an approximate mean and variance evaluation is useful involves signal in noise problems when the SNR is high. Then, the estimation performance can be partially described by the approximate mean of (9.18) and the approximate variance of (9.19).

9.3 Method of Moments

The method of moments approach to estimation is based on the solution of a theoretical equation involving the moments of a PDF. As an example, assume we observe $x[n]$ for $n = 0, 1, \ldots, N-1$, which are IID samples from the Gaussian mixture PDF (see also

Problem 6.14)

$$p(x[n];\epsilon) = \frac{1-\epsilon}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{1}{2}\frac{x^2[n]}{\sigma_1^2}\right) + \frac{\epsilon}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{1}{2}\frac{x^2[n]}{\sigma_2^2}\right)$$

or in more succinct form

$$p(x[n];\epsilon) = (1-\epsilon)\phi_1(x[n]) + \epsilon\phi_2(x[n])$$

where

$$\phi_i(x[n]) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{1}{2}\frac{x^2[n]}{\sigma_i^2}\right).$$

The parameter ϵ is termed the mixture parameter, which satisfies $0 < \epsilon < 1$, and σ_1^2, σ_2^2 are the variances of the individual Gaussian PDFs. The Gaussian mixture PDF may be thought of as the PDF of a random variable obtained from a $\mathcal{N}(0,\sigma_1^2)$ PDF with probability $1-\epsilon$ and from a $\mathcal{N}(0,\sigma_2^2)$ PDF with probability ϵ. Now if σ_1^2, σ_2^2 are known and ϵ is to be estimated, all our usual MVU estimation methods will fail. The MLE will require the maximization of a very nonlinear function of ϵ. Although this can be implemented using a grid search, the method of moments provides a much simpler estimator. Note that

$$\begin{aligned} E(x^2[n]) &= \int_{-\infty}^{\infty} x^2[n]\left[(1-\epsilon)\phi_1(x[n]) + \epsilon\phi_2(x[n])\right] dx[n] \\ &= (1-\epsilon)\sigma_1^2 + \epsilon\sigma_2^2 \end{aligned} \tag{9.1}$$

since the mean of $x[n]$ is zero. This theoretical equation relates the unknown parameter ϵ to the second moment. If we now replace $E(x^2[n])$ by its natural estimator $\frac{1}{N}\sum_{n=0}^{N-1} x^2[n]$, we have

$$\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] = (1-\epsilon)\sigma_1^2 + \epsilon\sigma_2^2 \tag{9.2}$$

and solving for ϵ, we have our method of moments estimator

$$\hat{\epsilon} = \frac{\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] - \sigma_1^2}{\sigma_2^2 - \sigma_1^2}. \tag{9.3}$$

The ease with which the estimator was found is due to the linear nature of the theoretical moment equation (9.1). The estimator is easily shown to be unbiased (although this is not generally the case). The variance of $\hat{\epsilon}$ is found as

$$\begin{aligned} \text{var}(\hat{\epsilon}) &= \frac{1}{(\sigma_2^2-\sigma_1^2)^2}\text{var}\left(\frac{1}{N}\sum_{n=0}^{N-1} x^2[n]\right) \\ &= \frac{1}{N(\sigma_2^2-\sigma_1^2)^2}\text{var}(x^2[n]) \\ &= \frac{1}{N(\sigma_2^2-\sigma_1^2)^2}\left[E(x^4[n]) - E^2(x^2[n])\right]. \end{aligned}$$

But it is easily shown that

$$E(x^4[n]) = (1-\epsilon)3\sigma_1^4 + \epsilon 3\sigma_2^4$$

which when combined with (9.1) produces the variance

$$\text{var}(\hat{\epsilon}) = \frac{3(1-\epsilon)\sigma_1^4 + 3\epsilon\sigma_2^4 - [(1-\epsilon)\sigma_1^2 + \epsilon\sigma_2^2]^2}{N(\sigma_2^2 - \sigma_1^2)^2}. \tag{9.4}$$

To determine the loss in performance we could evaluate the CRLB (we would need to do this numerically) and compare it to (9.4). If the increase in variance were substantial, we could attempt to implement the MLE, which would attain the bound for large data records. It should be observed that the method of moments estimator is consistent in that $\hat{\epsilon} \to \epsilon$ as $N \to \infty$. This is because as $N \to \infty$, from (9.3) $E(\hat{\epsilon}) = \epsilon$ and from (9.4) $\text{var}(\hat{\epsilon}) \to 0$. Since the estimates of the moments that are substituted into the theoretical equation approach the true moments for large data records, the equation to be solved approaches the theoretical equation. As a result, the method of moments estimator will in general be consistent (see Problem 7.5). In (9.2) for example, as $N \to \infty$, we have

$$\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] \to E(x^2[n])$$

and the theoretical equation results. When solved for ϵ, the true value is obtained.

We now summarize the method of moments for a scalar parameter. Assume that the kth moment $\mu_k = E(x^k[n])$ depends upon the unknown parameter θ according to

$$\mu_k = h(\theta). \tag{9.5}$$

We first solve for θ as

$$\theta = h^{-1}(\mu_k)$$

assuming h^{-1} exists. We then replace the theoretical moment by its natural estimator

$$\hat{\mu}_k = \frac{1}{N}\sum_{n=0}^{N-1} x^k[n] \tag{9.6}$$

to yield the method of moments estimator

$$\hat{\theta} = h^{-1}\left(\frac{1}{N}\sum_{n=0}^{N-1} x^k[n]\right). \tag{9.7}$$

Some examples follow.

Example 9.1 - DC Level in WGN

If $x[n] = A + w[n]$ is observed for $n = 0, 1, \ldots, N-1$, where $w[n]$ is WGN with variance σ^2 and A is to be estimated, then we know that

$$\mu_1 = E(x[n]) = A.$$

This is the theoretical equation of (9.5). According to (9.7), we replace μ_1 by its natural estimator, resulting in

$$\hat{A} = \frac{1}{N}\sum_{n=0}^{N-1} x[n].$$

In this case h is the identity transformation or $h(x) = x$. ◇

Example 9.2 - Exponential PDF

Consider N IID observations from the exponential PDF

$$p(x[n];\lambda) = \begin{cases} \lambda \exp(-\lambda x[n]) & x[n] > 0 \\ 0 & x[n] < 0. \end{cases}$$

We wish to estimate the parameter λ, where $\lambda > 0$. The first moment is

$$\begin{aligned} \mu_1 = E(x[n]) &= \int_0^\infty x[n]\lambda \exp(-\lambda x[n])\, dx[n] \\ &= \frac{1}{\lambda}\int_0^\infty \xi \exp(-\xi)\, d\xi \\ &= \frac{1}{\lambda}. \end{aligned}$$

Solving for λ

$$\lambda = \frac{1}{\mu_1}$$

and substituting the natural estimator result in the method of moments estimator

$$\hat{\lambda} = \frac{1}{\frac{1}{N}\sum_{n=0}^{N-1} x[n]}.$$

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9.4 Extension to a Vector Parameter

Now consider a vector parameter $\boldsymbol{\theta}$ of dimension $p \times 1$. It is obvious that to solve for $\boldsymbol{\theta}$ requires p theoretical moment equations. Hence, we suppose

$$\begin{aligned} \mu_1 &= h_1(\theta_1, \theta_2, \ldots, \theta_p) \\ \mu_2 &= h_2(\theta_1, \theta_2, \ldots, \theta_p) \\ &\vdots \\ \mu_p &= h_p(\theta_1, \theta_2, \ldots, \theta_p) \end{aligned} \tag{9.8}$$

or in matrix form

$$\boldsymbol{\mu} = \mathbf{h}(\boldsymbol{\theta}). \tag{9.9}$$

We then solve for $\boldsymbol{\theta}$ as

$$\boldsymbol{\theta} = \mathbf{h}^{-1}(\boldsymbol{\mu}) \tag{9.10}$$

and determine the method of moments estimator as

$$\hat{\boldsymbol{\theta}} = \mathbf{h}^{-1}(\hat{\boldsymbol{\mu}}) \tag{9.11}$$

where

$$\hat{\boldsymbol{\mu}} = \begin{bmatrix} \frac{1}{N}\sum_{n=0}^{N-1} x[n] \\ \frac{1}{N}\sum_{n=0}^{N-1} x^2[n] \\ \vdots \\ \frac{1}{N}\sum_{n=0}^{N-1} x^p[n] \end{bmatrix}.$$

It may occur that the first p moments are insufficient to determine all the parameters to be estimated (see Example 9.3). In this case we need to find some set of p moment equations allowing (9.9) to be solved for $\boldsymbol{\theta}$ to obtain (9.10). In practice, it is desirable to use the lowest order moments possible. This is because the variance of the moment estimator generally increases with order. Also, to be able to solve the resulting equations we would like them to be *linear* or at least mildly nonlinear. Otherwise, a nonlinear optimization may be needed, defeating the original motivation for the method of moments estimator as an easily implemented estimator. In the vector parameter case we may also need cross-moments, as the signal processing example in Section 9.6 illustrates (see also Problem 9.5). We now continue the Gaussian mixture example.

Example 9.3 - Gaussian Mixture PDF

Returning to the introductory example of a Gaussian mixture PDF, assume now that in addition to ϵ, the Gaussian variances σ_1^2 and σ_2^2 are unknown as well. To estimate all three parameters we require three moment equations. Noting that the PDF is an even function so that all odd order moments are zero, we utilize

$$\begin{aligned} \mu_2 &= E(x^2[n]) = (1-\epsilon)\sigma_1^2 + \epsilon\sigma_2^2 \\ \mu_4 &= E(x^4[n]) = 3(1-\epsilon)\sigma_1^4 + 3\epsilon\sigma_2^4 \\ \mu_6 &= E(x^6[n]) = 15(1-\epsilon)\sigma_1^6 + 15\epsilon\sigma_2^6. \end{aligned}$$

Although nonlinear, these equations may be solved by letting [Rider 1961]

$$\begin{aligned} u &= \sigma_1^2 + \sigma_2^2 \\ v &= \sigma_1^2\sigma_2^2. \end{aligned} \tag{9.12}$$

Then it can be shown by direct substitution that

$$\begin{aligned} u &= \frac{\mu_6 - 5\mu_4\mu_2}{5\mu_4 - 15\mu_2^2} \\ v &= \mu_2 u - \frac{\mu_4}{3}. \end{aligned}$$

Once u is found, v can be determined. Then, σ_1^2 and σ_2^2 are obtained by solving (9.12) to yield

$$\begin{aligned} \sigma_1^2 &= \frac{u + \sqrt{u^2 - 4v}}{2} \\ \sigma_2^2 &= \frac{v}{\sigma_1^2} \end{aligned}$$

and finally, the mixture parameter becomes (see (9.3))

$$\epsilon = \frac{\mu_2 - \sigma_1^2}{\sigma_2^2 - \sigma_1^2}.$$

The method of moments performs the same operations but with the theoretical moments $\{\mu_2, \mu_4, \mu_6\}$ replaced by their natural estimators.

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9.5 Statistical Evaluation of Estimators

In the method of moments we do not know beforehand whether the estimator will perform well. Once the estimator has been obtained we may be able, if we are lucky, to determine the statistical properties of mean and variance as in the introductory Gaussian mixture example. Since

$$\hat{\boldsymbol{\theta}} = \mathbf{h}^{-1}(\hat{\boldsymbol{\mu}}) = \mathbf{g}(\mathbf{x}) \tag{9.13}$$

we can in principle determine the PDF of $\hat{\boldsymbol{\theta}}$ using standard formulas for the transformation of random variables. In practice, this is usually impossible due to its mathematical intractibility. Because (9.13) is the general form for any estimator, the methods to be described apply to other estimators in addition to the method of moments estimator. They allow us to assess the performance of an estimator by determining *approximate* expressions for the mean and variance. The approximation generally improves as the data record length increases, being an asymptotic approach. For determination of the *exact* mean and variance a Monte Carlo computer simulation as discussed in Chapter 7 must be employed.

Consider a scalar parameter that has been estimated using (9.13). To determine the approximate mean and variance of $\hat{\theta}$ we must assume that it depends upon the $r < N$ statistics $\{T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_r(\mathbf{x})\}$, whose variances and covariances are small. With

this latter assumption the PDF of $[T_1\, T_2 \ldots T_r]^T$ will be concentrated about its mean. In Example 9.2, for instance, the estimator of λ can be written as

$$\hat{\lambda} = g(T_1(\mathbf{x}))$$

where $T_1(\mathbf{x}) = \frac{1}{N}\sum_{n=0}^{N-1} x[n]$ and $g(T_1) = 1/T_1$. For large N the PDF of T_1 will be heavily concentrated about its mean since $\text{var}(T_1) = 1/(N\lambda^2)$, as will be shown in Example 9.4. Using the statistical linearization argument of Chapter 3, we can use a first-order Taylor expansion of g about the mean of T_1. In general, we assume that

$$\hat{\theta} = g(\mathbf{T})$$

where $\mathbf{T} = [T_1\, T_2 \ldots T_r]^T$. We then perform a first-order Taylor expansion of g about the point $\mathbf{T} = E(\mathbf{T}) = \boldsymbol{\mu}$ to yield

$$\hat{\theta} = g(\mathbf{T}) \approx g(\boldsymbol{\mu}) + \sum_{k=1}^{r} \left.\frac{\partial g}{\partial T_k}\right|_{\mathbf{T}=\boldsymbol{\mu}} (T_k - \mu_k). \tag{9.14}$$

Assuming this to hold, the mean becomes

$$E(\hat{\theta}) = g(\boldsymbol{\mu}). \tag{9.15}$$

To this degree of approximation $E(\hat{\theta}) = E(g(\mathbf{T})) = g(E(\mathbf{T}))$ or the expectation commutes over the nonlinear function g. Note that we require the first-order moment of $\mathbf{T}$ to determine the mean. To determine the variance we again use (9.14) to obtain

$$\text{var}(\hat{\theta}) = E\left\{\left[g(\boldsymbol{\mu}) + \left.\frac{\partial g}{\partial \mathbf{T}}\right|^{T}_{\mathbf{T}=\boldsymbol{\mu}} (\mathbf{T} - \boldsymbol{\mu}) - E(\hat{\theta})\right]^2\right\}.$$

But from (9.15) we have that $E(\hat{\theta}) = g(\boldsymbol{\mu})$. Therefore,

$$\text{var}(\hat{\theta}) = \left.\frac{\partial g}{\partial \mathbf{T}}\right|^{T}_{\mathbf{T}=\boldsymbol{\mu}} \mathbf{C}_T \left.\frac{\partial g}{\partial \mathbf{T}}\right|_{\mathbf{T}=\boldsymbol{\mu}} \tag{9.16}$$

where $\mathbf{C}_T$ is the covariance matrix of $\mathbf{T}$. Now we require both the mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{C}_T$ to determine the variance. Similar derivations can be used to extend the Taylor expansion to higher order terms, although the algebra becomes increasingly tedious (see Problems 9.8 and 9.9). An example follows.

Example 9.4 - Exponential PDF (continued)

Continuing Example 9.2, suppose we have the method of moments estimator

$$\hat{\lambda} = \frac{1}{\frac{1}{N}\sum_{n=0}^{N-1} x[n]}$$

where the $x[n]$'s are IID and each has an exponential distribution. To find the approximate mean and variance we use (9.15) and (9.16). In this case we have

$$\hat{\lambda} = g(T_1)$$

where

$$T_1 = \frac{1}{N}\sum_{n=0}^{N-1} x[n]$$

and

$$g(T_1) = \frac{1}{T_1}.$$

The mean of T_1 is

$$\begin{aligned} \mu_1 = E(T_1) &= \frac{1}{N}\sum_{n=0}^{N-1} E(x[n]) \\ &= E(x[n]) \\ &= \frac{1}{\lambda} \end{aligned}$$

using the results of Example 9.2. The variance of T_1 is

$$\begin{aligned} \mathrm{var}(T_1) &= \mathrm{var}\left(\frac{1}{N}\sum_{n=0}^{N-1} x[n]\right) \\ &= \frac{\mathrm{var}(x[n])}{N}. \end{aligned}$$

But

$$\begin{aligned} \mathrm{var}(x[n]) &= \int_0^\infty x^2[n]\lambda \exp(-\lambda x[n])\, dx[n] - \frac{1}{\lambda^2} \\ &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} \end{aligned}$$

so that

$$\mathrm{var}(T_1) = \frac{1}{N\lambda^2}.$$

Since $g(T_1) = 1/T_1$,

$$\left.\frac{\partial g}{\partial T_1}\right|_{T_1=\mu_1} = -\frac{1}{\mu_1^2} = -\lambda^2.$$

From (9.15) we have the approximate mean

$$E(\hat{\lambda}) = g(\mu_1) = \frac{1}{\frac{1}{\lambda}} = \lambda$$

and from (9.16) the approximate variance is

$$\begin{aligned} \text{var}(\hat{\lambda}) &= \left.\frac{\partial g}{\partial T_1}\right|_{T_1=\mu_1} \text{var}(T_1) \left.\frac{\partial g}{\partial T_1}\right|_{T_1=\mu_1} \\ &= (-\lambda^2)\frac{1}{N\lambda^2}(-\lambda^2) \\ &= \frac{\lambda^2}{N}. \end{aligned}$$

The estimator is seen to be approximately unbiased and has an approximate variance decreasing with N. It should be emphasized that these expressions are only approximate. They are only as accurate as is the linearization of g. In fact, for this problem $\hat{\lambda}$ can be easily shown to be the MLE. As such, its asymptotic PDF (as $N \to \infty$) can be shown to be (see (7.8))

$$\hat{\lambda} \overset{a}{\sim} \mathcal{N}(\lambda, \lambda^2/N).$$

Hence, the approximation is accurate only as $N \to \infty$. (In Problem 9.10 *quadratic* expansions of g are employed to derive the *second-order* approximations to the mean and variance of $\hat{\lambda}$.) To determine how large N must be for the mean and variance expressions to hold requires a Monte Carlo computer simulation. ◇

The basic premise behind the Taylor series approach is that the function g be approximately linear over the range of $\mathbf{T}$ for which $p(\mathbf{T};\theta)$ is essentially nonzero. This occurs naturally when

1. The data record is large so that $p(\mathbf{T};\theta)$ is concentrated about its mean, as in the previous example. There the PDF of $T_1 = \frac{1}{N}\sum_{n=0}^{N-1} x[n]$ becomes more concentrated about $E(T_1)$ as $N \to \infty$ since $\text{var}(T_1) = 1/(N\lambda^2) \to 0$ as $N \to \infty$.

2. The problem is to estimate the parameter of a signal in noise, and the SNR is high. In this case, by expanding the function about the signal we can obtain the approximate mean and variance. For a high SNR the results will be quite accurate. This is because at a high SNR the noise causes only a slight perturbation of the estimator value obtained in the case of no noise. We now develop this second case.

We consider the data

$$x[n] = s[n;\theta] + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is zero mean noise with covariance matrix $\mathbf{C}$. A general estimator of the scalar parameter θ is

$$\begin{aligned} \hat{\theta} &= g(\mathbf{x}) \\ &= g(\mathbf{s}(\theta) + \mathbf{w}) \\ &= h(\mathbf{w}). \end{aligned}$$

In this case we may choose the statistic $\mathbf{T}$ as the original data since as the SNR becomes larger, the PDF of $\mathbf{T} = \mathbf{x}$ becomes more concentrated about its mean, which is the signal. We now use a first-order Taylor expansion of g about the mean of $\mathbf{x}$, which is $\boldsymbol{\mu} = \mathbf{s}(\theta)$, or, equivalently, of h about $\mathbf{w} = \mathbf{0}$. This results in

$$\hat{\theta} \approx h(\mathbf{0}) + \sum_{n=0}^{N-1} \left. \frac{\partial h}{\partial w[n]} \right|_{\mathbf{w}=\mathbf{0}} w[n]. \tag{9.17}$$

As before, it follows that approximately

$$E(\hat{\theta}) = \mathbf{h}(\mathbf{0}) = g(\mathbf{s}(\theta)) \tag{9.18}$$

$$\text{var}(\hat{\theta}) = \left. \frac{\partial h}{\partial \mathbf{w}} \right|_{\mathbf{w}=\mathbf{0}}^{T} \mathbf{C} \left. \frac{\partial h}{\partial \mathbf{w}} \right|_{\mathbf{w}=\mathbf{0}}. \tag{9.19}$$

In most cases if there is no noise, the estimator yields the true value since $g(\mathbf{s}(\theta)) = \theta$. Hence, for a high SNR it will be unbiased. An example follows.

Example 9.5 - Exponential Signal in White Noise

For the data

$$x[n] = r^n + w[n] \qquad n = 0, 1, 2$$

where $w[n]$ is zero mean uncorrelated noise with variance σ^2, the damping factor r is to be estimated. Wishing to avoid a maximization of the likelihood to find the MLE (see Example 7.11), we propose the estimator

$$\hat{r} = \frac{x[2] + x[1]}{x[1] + x[0]}.$$

In terms of the signal and noise we have

$$\hat{r} = h(\mathbf{w}) = \frac{r^2 + w[2] + r + w[1]}{r + w[1] + 1 + w[0]}$$

and according to (9.18)

$$E(\hat{r}) = h(\mathbf{0}) = \frac{r^2 + r}{r + 1} = r$$

or the estimator is approximately unbiased. To find the variance

$$\begin{aligned} \left. \frac{\partial h}{\partial w[0]} \right|_{\mathbf{w}=\mathbf{0}} &= -\left. \frac{r^2 + w[2] + r + w[1]}{(r + w[1] + 1 + w[0])^2} \right|_{\mathbf{w}=\mathbf{0}} \\ &= -\frac{r^2 + r}{(r+1)^2} \\ &= -\frac{r}{r+1}. \end{aligned}$$

Similarly,

$$\begin{aligned} \left.\frac{\partial h}{\partial w[1]}\right|_{\mathbf{w}=\mathbf{0}} &= -\frac{r-1}{r+1} \\ \left.\frac{\partial h}{\partial w[2]}\right|_{\mathbf{w}=\mathbf{0}} &= \frac{1}{r+1}. \end{aligned}$$

Since $\mathbf{C} = \sigma^2\mathbf{I}$, we have from (9.19)

$$\begin{aligned} \text{var}(\hat{r}) &= \sigma^2 \sum_{n=0}^{2} \left(\left.\frac{\partial h}{\partial w[n]}\right|_{\mathbf{w}=\mathbf{0}} \right)^2 \\ &= \frac{\sigma^2}{(r+1)^2}[r^2 + (r-1)^2 + 1] \\ &= 2\sigma^2 \frac{r^2 - r + 1}{(r+1)^2}. \end{aligned}$$

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It is also worthwhile to remark that from (9.14) and (9.17) the estimators are approximately linear functions of $\mathbf{T}$ and $\mathbf{w}$, respectively. If these are Gaussian, then $\hat{\theta}$ will also be Gaussian — at least to within the approximation assumed by the Taylor expansion. Finally, we should note that for a *vector parameter* the approximate mean and variance for each component can be obtained by applying these techniques. Obtaining the *covariance matrix* of $\hat{\boldsymbol{\theta}}$ is possible using a first-order Taylor expansion but can be extremely tedious.

9.6 Signal Processing Example

We now apply the method of moments and the approximate performance analysis to the problem of frequency estimation. Assume that we observe

$$x[n] = A\cos(2\pi f_0 n + \phi) + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is zero mean white noise with variance σ^2. The frequency f_0 is to be estimated. This problem was discussed in Example 7.16 in which the MLE of frequency was shown to be approximately given by the peak location of a periodogram. In an effort to reduce the computation involved in searching for the peak location, we now describe a method of moments estimator. To do so we depart slightly from the usual sinusoidal model to assume that the phase ϕ is a random variable independent of $w[n]$ and distributed as $\phi \sim \mathcal{U}[0, 2\pi]$. With this assumption the signal $s[n] = A\cos(2\pi f_0 n + \phi)$ can be viewed as the *realization* of a WSS random process. That this is the case is verified by determining the mean and ACF of $s[n]$. The mean is

$$\begin{aligned} E(s[n]) &= E[A\cos(2\pi f_0 n + \phi)] \\ &= \int_0^{2\pi} A\cos(2\pi f_0 n + \phi)\frac{1}{2\pi}\,d\phi \\ &= 0 \end{aligned}$$

and the ACF is

$$\begin{aligned} r_{ss}[k] &= E[s[n]s[n+k]] \\ &= E[A^2\cos(2\pi f_0 n + \phi)\cos(2\pi f_0(n+k) + \phi)] \\ &= A^2 E\left[\frac{1}{2}\cos(4\pi f_0 n + 2\pi f_0 k + 2\phi) + \frac{1}{2}\cos 2\pi f_0 k\right] \\ &= \frac{A^2}{2}\cos 2\pi f_0 k. \end{aligned}$$

The ACF of the observed process becomes

$$\begin{aligned} r_{xx}[k] &= r_{ss}[k] + r_{ww}[k] \\ &= \frac{A^2}{2}\cos 2\pi f_0 k + \sigma^2\delta[k]. \end{aligned}$$

To simplify the discussion and the subsequent estimator we will assume that the signal amplitude is known (see Problem 9.12 for the unknown amplitude case). We will let $A = \sqrt{2}$, so that the ACF becomes

$$r_{xx}[k] = \cos 2\pi f_0 k + \sigma^2\delta[k].$$

To estimate the frequency by the method of moments approach we observe that

$$r_{xx}[1] = \cos 2\pi f_0,$$

and therefore without having to assume knowledge of σ^2 we can implement the method of moments estimator

$$\hat{f}_0 = \frac{1}{2\pi}\arccos \hat{r}_{xx}[1]$$

where $\hat{r}_{xx}[1]$ is an estimator of $r_{xx}[1]$. A reasonable estimator of the ACF for $k = 1$ is

$$\hat{r}_{xx}[1] = \frac{1}{N-1}\sum_{n=0}^{N-2} x[n]x[n+1]$$

so that finally we have for our frequency estimator

$$\hat{f}_0 = \frac{1}{2\pi}\arccos\left[\frac{1}{N-1}\sum_{n=0}^{N-2} x[n]x[n+1]\right]. \tag{9.20}$$

The argument of the arccos function may exceed 1 in magnitude, usually occurring when the SNR is low. If this is the case, the estimate is meaningless and it is indicative of an unreliable estimator.

Although motivated by the random phase sinusoidal model, this estimator remains valid if ϕ is deterministic and unknown (see Problem 9.11). In determining the mean and variance of $\hat{f}_0$ we assume that ϕ is deterministic. To assess the performance of our estimator we use the first-order Taylor expansion approach and expand about the point $\mathbf{w} = \mathbf{0}$. Thus, the obtained performance is valid for high enough SNR. From (9.18) we have

$$\begin{aligned} E(\hat{f}_0) &= \frac{1}{2\pi} \arccos \left[\frac{1}{N-1} \sum_{n=0}^{N-2} \sqrt{2}\cos(2\pi f_0 n + \phi)\sqrt{2}\cos(2\pi f_0(n+1) + \phi) \right] \\ &= \frac{1}{2\pi} \arccos \left[\frac{1}{N-1} \sum_{n=0}^{N-2} (\cos(4\pi f_0 n + 2\pi f_0 + 2\phi) + \cos 2\pi f_0) \right]. \end{aligned}$$

The double-frequency term when summed is approximately zero, so that at a high SNR

$$\begin{aligned} E(\hat{f}_0) &= \frac{1}{2\pi} \arccos [\cos 2\pi f_0] \\ &= f_0. \end{aligned} \tag{9.21}$$

To find the variance we note that

$$\hat{f}_0 = h(\mathbf{w}) = \frac{1}{2\pi} \arccos \left[\underbrace{\frac{1}{N-1} \sum_{n=0}^{N-2} (s[n] + w[n])(s[n+1] + w[n+1])}_{u} \right]$$

so that the first-order partials are

$$\left. \frac{\partial h}{\partial w[i]} \right|_{\mathbf{w}=\mathbf{0}} = -\frac{1}{2\pi} \frac{1}{\sqrt{1-u^2}} \left. \frac{\partial u}{\partial w[i]} \right|_{\mathbf{w}=\mathbf{0}}.$$

But

$$\left. \frac{\partial u}{\partial w[i]} \right|_{\mathbf{w}=\mathbf{0}} = \frac{1}{N-1} \frac{\partial}{\partial w[i]} \sum_{n=0}^{N-2} (s[n]w[n+1] + w[n]s[n+1]) \Bigg|_{\mathbf{w}=\mathbf{0}}$$

with the other terms in the sum resulting in zero after differentiating and letting $\mathbf{w} = \mathbf{0}$. Continuing, we have

$$\left. \frac{\partial u}{\partial w[i]} \right|_{\mathbf{w}=\mathbf{0}} = \begin{cases} \frac{1}{N-1} s[1] & i = 0 \\ \frac{1}{N-1} (s[i-1] + s[i+1]) & i = 1, 2, \ldots, N-2 \\ \frac{1}{N-1} s[N-2] & i = N-1 \end{cases}$$

Also,

$$\begin{aligned} u|_{\mathbf{w}=\mathbf{0}} &= \frac{1}{N-1}\sum_{n=0}^{N-2} s[n]s[n+1] \\ &\approx \cos 2\pi f_0 \end{aligned}$$

as already shown. Thus,

$$\left.\frac{\partial h}{\partial w[i]}\right|_{\mathbf{w}=\mathbf{0}} = -\frac{1}{2\pi \sin 2\pi f_0}\left.\frac{\partial u}{\partial w[i]}\right|_{\mathbf{w}=\mathbf{0}}.$$

From (9.19) with $\mathbf{C} = \sigma^2\mathbf{I}$ we obtain

$$\begin{aligned} \mathrm{var}(\hat{f}_0) &= \sigma^2 \sum_{n=0}^{N-1} \left(\left.\frac{\partial h}{\partial w[n]}\right|_{\mathbf{w}=\mathbf{0}}\right)^2 \\ &= \frac{\sigma^2}{(2\pi)^2(N-1)^2\sin^2 2\pi f_0}\left[s^2[1] \right. \\ &\qquad \left. + \sum_{n=1}^{N-2}(s[n-1]+s[n+1])^2 + s^2[N-2]\right]. \end{aligned}$$

But $s[n-1]+s[n+1] = 2\cos 2\pi f_0 s[n]$, as can easily be verified, so that finally the variance of the frequency estimator is for a high SNR

$$\mathrm{var}(\hat{f}_0) = \frac{\sigma^2}{(2\pi)^2(N-1)^2\sin^2 2\pi f_0}\left[s^2[1] + 4\cos^2 2\pi f_0 \sum_{n=1}^{N-2} s^2[n] + s^2[N-2]\right] \quad (9.22)$$

where $s[n] = \sqrt{2}\cos(2\pi f_0 n + \phi)$. The evaluation of the variance is straightforward, although tedious. Note that the variance decreases most rapidly as $1/N^2$ (for $f_0 = 1/4$) versus the $1/N^3$ dependence for the CRLB (see (3.41)). Additionally, for f_0 near 0 or $1/2$ the variance increases rapidly. This is due to the arccos operation required to determine $\hat{f}_0$. As shown in Figure 9.1, a slight perturbation in the argument for x near ± 1 (f_0 near 0 or $1/2$) will cause large fluctuations in the function value due to the steep slope.

As an example of the accuracy of the approximate mean and variance expressions, we consider the data set with parameters SNR $= A^2/2\sigma^2 = 1/\sigma^2$ for $A = \sqrt{2}$ and $f_0 = 0.2$, $\phi = 0$, and $N = 50$. For $w[n]$ being WGN a Monte Carlo computer simulation in which 1000 realizations of $\hat{f}_0$ as given by (9.20) were generated for each of several SNRs. In Figure 9.2a the actual mean $E(\hat{f}_0)$ obtained, as well as the theoretical mean of $f_0 = 0.2$, are shown versus SNR. As expected for a high enough SNR, the estimator is essentially unbiased. The slight discrepancy is due to the approximations made in deriving the estimator, principally that N was large enough to neglect the double-frequency term. For larger N it can be verified that the mean equals the true value

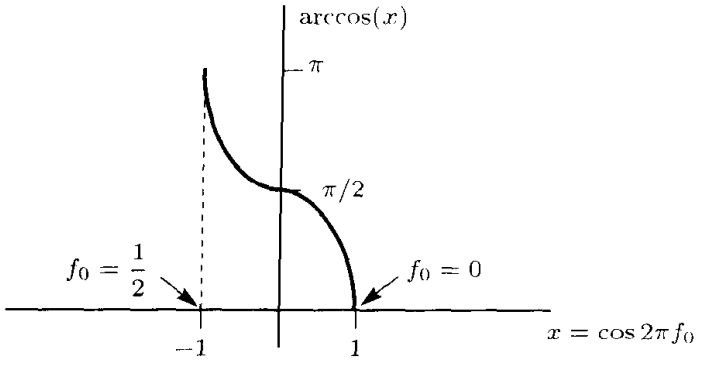
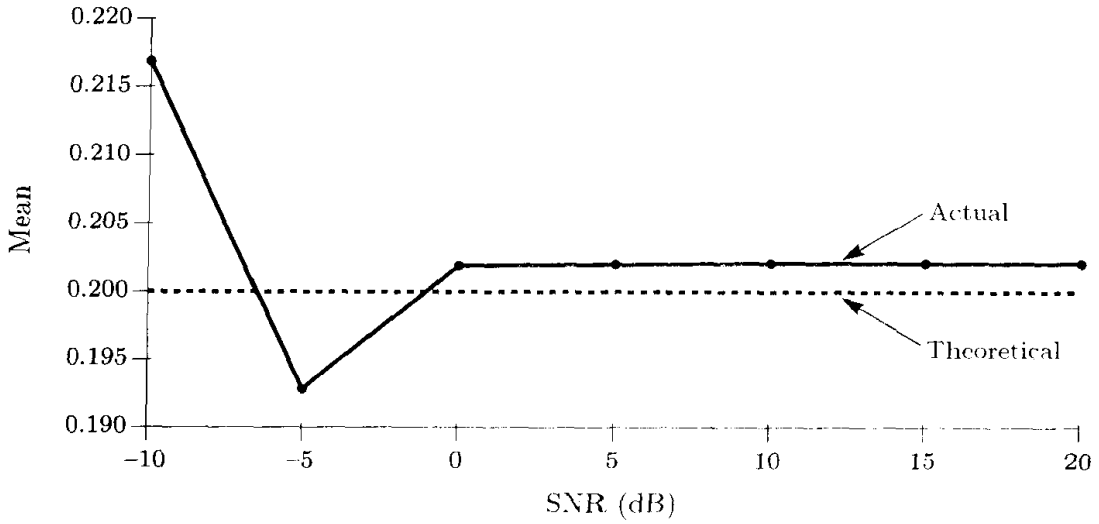
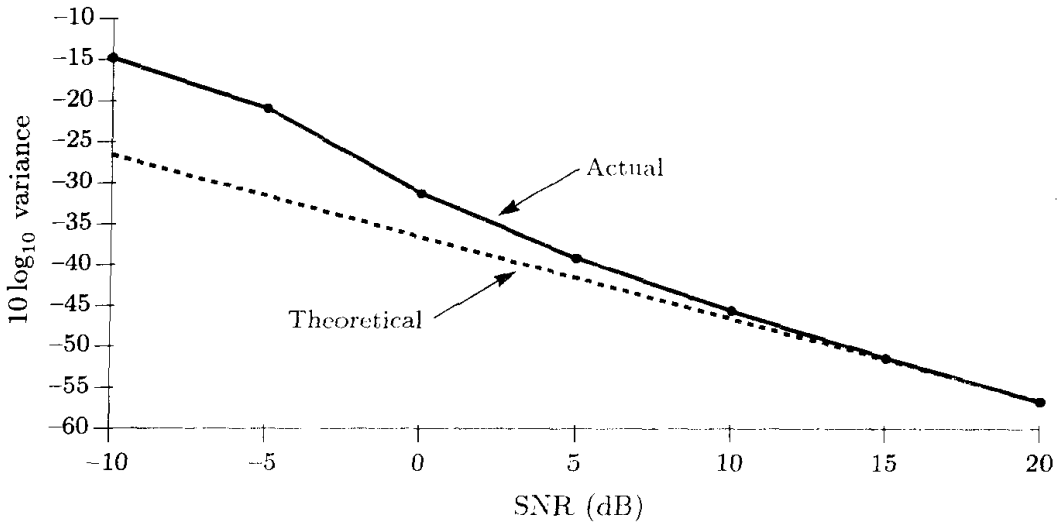


Figure 9.1 Explanation for increased variance of frequency estimator



(a)



(b)

Figure 9.2 Method of moments frequency estimator

for a high SNR. Correspondingly, in Figure 9.2b the actual variance obtained and the theoretical variance as determined from (9.22) are shown versus SNR. The same behavior is evident. Some further discussions of this example can be found in Problem 9.13. For further discussions of this estimation approach the reader should consult [Lank et al. 1973, Kay 1989].

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Problems

9.1 If N IID observations $\{x[0], x[1], \ldots, x[N-1]\}$ are made from the Rayleigh PDF

$$p(x;\sigma^2) = \begin{cases} \frac{x}{\sigma^2} \exp\left(-\frac{1}{2}\frac{x^2}{\sigma^2}\right) & x > 0 \\ 0 & x < 0 \end{cases}$$

find a method of moments estimator for σ^2.

9.2 If N IID observations $\{x[0], x[1], \ldots, x[N-1]\}$ are made from the Laplacian PDF

$$p(x;\sigma) = \frac{1}{\sqrt{2}\sigma} \exp\left(-\frac{\sqrt{2}|x|}{\sigma}\right)$$

find a method of moments estimator for σ.

9.3 Assume that N IID samples from a bivariate Gaussian PDF are observed or $\{\mathbf{x}_0, \mathbf{x}_1, \ldots \mathbf{x}_{N-1}\}$, where each $\mathbf{x}$ is a 2×1 random vector with PDF $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$. If

$$\mathbf{C} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$$

find a method of moments estimator for ρ. Also, determine a cubic equation to be solved for the MLE of ρ. Comment on the ease of implementation of the different estimators.

9.4 If N IID observations $\{x[0], x[1], \ldots, x[N-1]\}$ are made from the $\mathcal{N}(\mu, \sigma^2)$ PDF, find a method of moments estimator for $\boldsymbol{\theta} = [\mu\ \sigma^2]^T$.

9.5 In Example 8.12 we determined that the ACF of an ARMA process satisfies the recursive difference equation

$$r_{xx}[n] = -\sum_{k=1}^{p} a[k] r_{xx}[n-k] \qquad n > q$$

where $\{a[1], a[2], \ldots, a[p]\}$ denote the AR filter parameters and q is the MA order. Propose a method of moments estimator for the AR filter parameters.

9.6 For a DC level in WGN or

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2, the parameter A^2 is to be estimated. It is proposed to use

$$\widehat{A^2} = (\bar{x})^2.$$

For this estimator find the approximate mean and variance using a first-order Taylor expansion approach.

9.7 For the observed data

$$x[n] = \cos\phi + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2, find a method of moments estimator for ϕ. Assuming the SNR is high, determine the approximate mean and variance for your estimator using the first-order Taylor expansion approach.

9.8 In this problem we determine the mean of an estimator to *second order* by using a second-order Taylor expansion. To do so expand $\hat{\theta} = g(\mathbf{T})$ about the mean of $\mathbf{T}$ or $\mathbf{T} = \boldsymbol{\mu}$, retaining the terms up to and including the quadratic term. Show that the approximate mean of $\hat{\theta}$ is

$$E(\hat{\theta}) = g(\boldsymbol{\mu}) + \frac{1}{2}\mathrm{tr}[\mathbf{G}(\boldsymbol{\mu})\mathbf{C}_T]$$

where

$$[\mathbf{G}(\boldsymbol{\mu})]_{ij} = \left.\frac{\partial^2 g}{\partial T_i \partial T_j}\right|_{\mathbf{T}=\boldsymbol{\mu}}.$$

Hint: Use the result

$$E(\mathbf{x}^T\mathbf{y}) = E(\mathrm{tr}(\mathbf{y}\mathbf{x}^T)) = \mathrm{tr}(E(\mathbf{y}\mathbf{x}^T)).$$

9.9 In this problem we determine the approximate variance of an estimator to *second order* or by using a second-order Taylor expansion approach. To do so first expand $\hat{\theta} = g(\mathbf{T})$ about the mean of $\mathbf{T}$ or $\mathbf{T} = \boldsymbol{\mu}$, retaining the terms up to and including

the quadratic term. Next, use the results of Problem 9.8 and finally make the assumption that $\mathbf{T} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C}_T)$. You should be able to verify the following result:

$$\text{var}(\hat{\theta}) = \left.\frac{\partial g}{\partial \mathbf{T}}\right|_{\mathbf{T}=\boldsymbol{\mu}}^{T} \mathbf{C}_T \left.\frac{\partial g}{\partial \mathbf{T}}\right|_{\mathbf{T}=\boldsymbol{\mu}} + \frac{1}{2}\text{tr}\left[(\mathbf{G}(\boldsymbol{\mu})\mathbf{C}_T)^2\right].$$

The $r \times r$ $\mathbf{G}(\boldsymbol{\mu})$ is defined in Problem 9.8. Hint: You will need the following result, which is valid for a symmetric $N \times N$ matrix $\mathbf{A}$,

$$\text{var}(\mathbf{x}^T\mathbf{A}\mathbf{x}) = 2\text{tr}\left[(\mathbf{A}\mathbf{C}_x)^2\right]$$

if $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_x)$.

9.10 Using the results of Problems 9.8 and 9.9 find the approximate mean and variance to second order for the estimator of λ discussed in Example 9.4. How do your results compare to the first-order result? Be sure to justify the approximate Gaussian PDF of $\bar{x}$, required to apply the variance expression. Also, compare the results to those predicted from asymptotic MLE theory (recall that the estimator is also the MLE).

9.11 Prove that for a sinusoidal signal

$$s[n] = A\cos(2\pi f_0 n + \phi)$$

where the phase ϕ is deterministic but unknown, that

$$\frac{1}{N-1}\sum_{n=0}^{N-2} s[n]s[n+1] \to \frac{A^2}{2}\cos 2\pi f_0$$

as $N \to \infty$. Assume that f_0 is not near 0 or 1/2. Hence, comment on the use of the method of moments estimator proposed in the signal processing example in Section 9.6 for the deterministic phase sinusoid.

9.12 To extend the applicability of the frequency estimator proposed in the signal processing example in Section 9.6 now assume that the signal amplitude A is unknown. Why can't the proposed estimator (9.20) be used in this case? Consider the estimator

$$\hat{f}_0 = \frac{1}{2\pi}\arccos\left[\frac{\frac{1}{N-1}\sum_{n=0}^{N-2} x[n]x[n+1]}{\frac{1}{N}\sum_{n=0}^{N-1} x^2[n]}\right].$$

Show that at a high SNR and for large N, $E(\hat{f}_0) = f_0$. Justify this estimator as a method of moments estimator based on the ACF. Would you expect this estimator to work well at a lower SNR?

9.13 For the signal processing example in Section 9.6 assume that the parameters are $A = \sqrt{2}, f_0 = 0.25, \phi = 0$, and that N is odd. Show that from (9.22) $\text{var}(\hat{f}_0) = 0$. To understand this result note that

$$\left.\frac{\partial h}{\partial w[i]}\right|_{\mathbf{w}=\mathbf{0}} = 0$$

for $i = 0, 1, \ldots, N-1$. What does this say about the first-order Taylor expansion approximation? How could you improve this situation?